

Capstone Project Overview

Overview

This program's capstone will reflect the concepts and technical knowledge you have acquired throughout the program, and it will showcase your ability to develop and answer a question of interest using the tools you have learned. Your final capstone project will focus on both the application of a predictive model and your ability to communicate your findings. By the end of Module 24, you will have completed the following:

- Build a predictive model using supervised or unsupervised learning techniques on a dataset of your choosing
- Develop a technical write-up in Jupyter Notebook posted on your GitHub repository
- Create a non-technical report describing your findings

Timeline/Deliverables

Module 6: Draft the Problem Statement

In Module 6, you will define your problem statement and develop a prospectus of the project. The prospectus provides a general overview of the question you will be asking, the data you require to answer the question, and the techniques you might use to answer the question.

Modules 11–14: First One-on-One Session with Your Program Leader

Sometime during these modules, you and your Program Leader will find a mutually agreeable time to talk synchronously about the problem statement you articulated in Module 6. Your Program Leader may:

- Help you refine your initial question and data needs
- Suggest where to source the data you will need
- Suggest methodologies for answering your question.

Remember that your Program Leader may propose unfamiliar methodologies that better serve your needs.

Module 16: Finalize the Problem Statement

In Module 16, you will submit your formal problem proposal, including:

- The research question you intend to answer
- Expected data sources and structure
- Expected results
- Expected techniques

Module 20: Initial Report and EDA

In Module 20, you will submit the following:

- A completed Jupyter notebook with your problem statement/question, data sources/structure, results, and techniques used (essentially, a fully built out version of what you submitted in Module 16)

Modules 21–23: Second One-on-One Session with Your Program Leader

After you submit your initial findings, you and your Program Leader will find a mutually agreeable time to go through your results to:

- Identify any gaps/errors in the findings
- Suggest possible further explorations (after the program)
- Answer questions about the non-technical report

Module 24: Final Analysis and Report

In Module 24, you will submit your non-technical report describing your capstone project. This report is designed to accompany your Jupyter notebook, providing a non-technical write-up that covers your problem, results, important findings, and suggestions for next steps.

To help you structure your report, we have provided a template for your reference, Final Report Template.

Capstone Project

{Insert project name} Final Report

1. **Define the Problem Statement:** Provide a well-defined problem statement that clearly articulates the goals, challenges, and potential benefits of your machine learning solution.
2. **Model Outcomes or Predictions:** Identify the type of learning (classification or regression) and specify the expected output of your selected model. Determine whether supervised or unsupervised learning algorithms will be used.
3. **Data Acquisition:** The deliverable at this step is to identify what data you plan to acquire and use with your model. For the best results, data should come from multiple sources and your analysis for including specific data should be clear. Please provide a clear visualization to assess the data's potential to solve the problem as well.
4. **Data Preprocessing/Preparation:** For this deliverable, you are tasked with detailing how you cleaned the data for your notebook.
 - a. What techniques did you use to ensure your data was free of missing values, and inconsistencies?
 - b. How did you split the data into training and test sets?
 - c. Please include any necessary analysis and encoding steps you took as well.
5. **Modeling:** For this deliverable, please document your selection of machine learning algorithms that you selected for your problem statement from the first deliverable.
6. **Model Evaluation:** Share your model evaluation here. What types of models did you consider for your problem (classification, regression, unsupervised)? Articulate the evaluation metrics you used and how you determined which model was most optimal for your problem.